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Air Quality Forecasting Using Machine Learning: Comparative Analysis and Ensemble Strategies for Enhanced Prediction

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Abstract Air pollution poses a critical challenge to environmental sustainability, public health, and urban planning. Accurate air quality prediction is essential for devising effective management strategies and early warning systems. This study utilized a dataset comprising hourly measurements of pollutants such as PM2.5, NO_x, CO, and benzene, sourced from five metal oxide sensors and a certified analyzer in a polluted urban area, totaling 9,357 records collected over one year (March 2004–February 2005) from the Kaggle Air Quality Data Set. A comprehensive comparison of ten machine learning regression models XGBoost, LightGBM, Random Forest, Gradient Boosting, CatBoost, Support Vector Regression (SVR) with Bayesian Optimization, Decision Tree, K-Nearest Neighbors (KNN), Elastic Net, and Bayesian Ridge was conducted. Model performance was enhanced through Bayesian optimization and randomized cross-validation, with stacking employed to leverage the strengths of base models. Experimental results showed that hyperparameter optimization and ensemble strategies significantly improved accuracy, with the SVR model optimized via Bayesian optimization achieving the highest performance: an R^2 score of 99.94%, MAE of 0.0120, and MSE of 0.0005. These findings underscore the methodology's efficacy in precisely capturing the spatial and temporal dynamics of air pollution.

Keywords Air Quality Prediction · Machine Learning · Bayesian Optimization · Regression Models · SVR

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1 Introduction

Air pollution is a global issue affecting environmental sustainability, public health, and urban planning. Rising industrialization, population density, and motor vehicle use degrade air quality, particularly in cities, contributing to respiratory diseases, cardiovascular disorders, and chronic health conditions (Zhang et al., 2024). The World Health Organization (WHO) and European Environment Agency (EEA) have proposed measures to mitigate air pollution's health impacts. However, effective management and preventive policies require accurate prediction of its spatial and temporal variations (Zhang et al., 2024).

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Traditional air pollution prediction relies on deterministic models rooted in atmospheric chemistry and physical principles. Chemical Transport Models (CTMs) simulate pollution distribution using meteorological data and emission inventories but depend heavily on input accuracy and struggle with complex atmospheric processes. High computational costs also make large-scale simulations time-consuming and expensive (Anggraini et al., 2024). In contrast, machine learning (ML) and deep learning (DL) approaches have emerged as flexible, data-driven alternatives for air quality prediction (Yu et al., 2025).

This study evaluates ten regression models for air quality prediction: XGBoost, LightGBM, Random Forest, Gradient Boosting, CatBoost, SVR, Decision Tree, KNN, Elastic Net, and Bayesian Ridge. Its primary goal is to compare their performance and identify optimal model structures (Yu et al., 2025). Hyperparameter optimization is critical for improving model accuracy. Typically set through trial-and-error or heuristic methods, hyperparameters are time-consuming to tune and may not yield optimal results (Lin et al., 2025). This study employs Bayesian Optimization and Randomized Cross-Validation (CV) to rigorously adjust hyperparameters, with Bayesian Optimization offering an efficient search process via probabilistic modeling (Lin et al., 2025). Stacking models combine the strengths of base models to better capture complex data relationships (Nandi et al., 2024). For example, gradient-boosted models like XGBoost and LightGBM deliver high accuracy, while SVR and Bayesian Ridge provide balanced predictions. Stacking integrates these advantages, enhancing prediction accuracy (Ahmed et al., 2024).

The literature underscores the potential of ML and DL for air quality prediction. Mendez et al. (2023) reviewed 155 studies from 2011–2021, analyzing ML/DL models' geographical distribution (Asia, Europe), parameters (PM2.5, NO_2), and algorithms, highlighting time-series models' effectiveness and the need for explainable AI (Méndez et al., 2023). Gupta et al. (2023) predicted AQI across cities using SVR and Random Forest, with SMOTE-balanced data yielding low Root Mean Square Error (RMSE) for Random Forest (Gupta et al., 2023). Guo et al. (2025) used DRL for HVAC systems, achieving 21.4% energy savings and better indoor air quality (Guo et al., 2025). Kothandaraman et al. (2022) found XGBoost and AdaBoost effective for PM2.5 prediction (Kothandaraman, et al., 2022). Mampitiya et al. (2023) reported high accuracy with LightGBM for PM10 prediction in Sri Lanka (Mampitiya, et al., 2023). Rybarczyk and Zalakeviciute (Rybarczyk & Zalakeviciute, 2021) noted reductions in NO_2, SO_2, CO, and PM2.5 during Quito's lockdown (Rybarczyk & Zalakeviciute, 2021). Wang et al. (2024) achieved accurate CO predictions in Nanjing using Convolutional Neural Network (CNN) (Wang et al., 2024). Liu et al. (2024) demonstrated LightGBM and LSTM's effectiveness for air quality prediction (Liu et al., 2024). Meena et al. (2024) linked air pollution to travel preferences (Meena et al., 2024). Rahman et al. (2024) proposed early-detection systems (Rahman et al., 2024). Ansari and Quaff (2025) predicted AQI in India (Ansari & Quaff, 2025). Wang and Zhang (2025) highlighted CNN's superiority (Wang & Zhang, 2025). Jiang et al. (2025) analyzed land use impacts (Jiang et al., 2025).

While many studies focus on limited models, this study comprehensively assesses XGBoost, LightGBM, Random Forest, Gradient Boosting, CatBoost, SVR, Decision Tree, KNN, Elastic Net, and Bayesian Ridge. Bayesian Optimization and Randomized CV minimize overfitting, achieving high accuracy. The optimized SVR model, for instance, recorded an Coefficient of Determination (R^2) of 0.9994, Mean Absolute Error (MAE) of 0.0120, and Mean Squared Error (MSE) of 0.0005. Stacking integrates individual model strengths, capturing complex data relationships effectively. This study offers a unique contribution by analyzing air quality's spatial and temporal dynamics with high accuracy.

2 Material and Method

This study aims to predict air pollution dynamics using machine learning models and to reveal complex relationships in pollutant concentrations. The data set provided includes basic pollutants such as PM2.5, NO_2, CO as well as air quality indices, and statistical consistency is ensured by outlier removal and correlation analysis in the data pre-processing stage. Ten different regression models such as XGBoost, LightGBM, Random Forest are trained with hyperparameter optimization and the results of these models are combined with the stack ensemble method to optimize the prediction accuracy. These results provide

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an effective methodological framework for practical applications such as air quality management and early warning systems, while providing a data-driven basis for policies to control pollutant sources. Figure 1 shows the flow diagram of the models used.

Hyperparameter optimization to improve the performance of machine learning models is at the heart of this work. Bayesian optimization and randomized CV methods are used to optimize the balance between complexity and generalization of the models and to minimize the risk of overlearning. In the study, 10 different regression models were trained: XGBoost, LightGBM, Random Forest, Gradient Boosting, CatBoost, SVR, Decision Tree, KNN, Elastic Net and Bayesian Ridge. Each of these models was configured to predict pollutant concentrations in the air quality dataset after hyperparameter optimization. The outputs of these models were then combined using a stack ensemble method to produce final predictions from a meta-model. The stack ensemble approach aimed to overcome the limitations of a single model by synthesizing the strengths of the base models and maximizing prediction accuracy. In particular, the use of algorithms such as linear regression or gradient

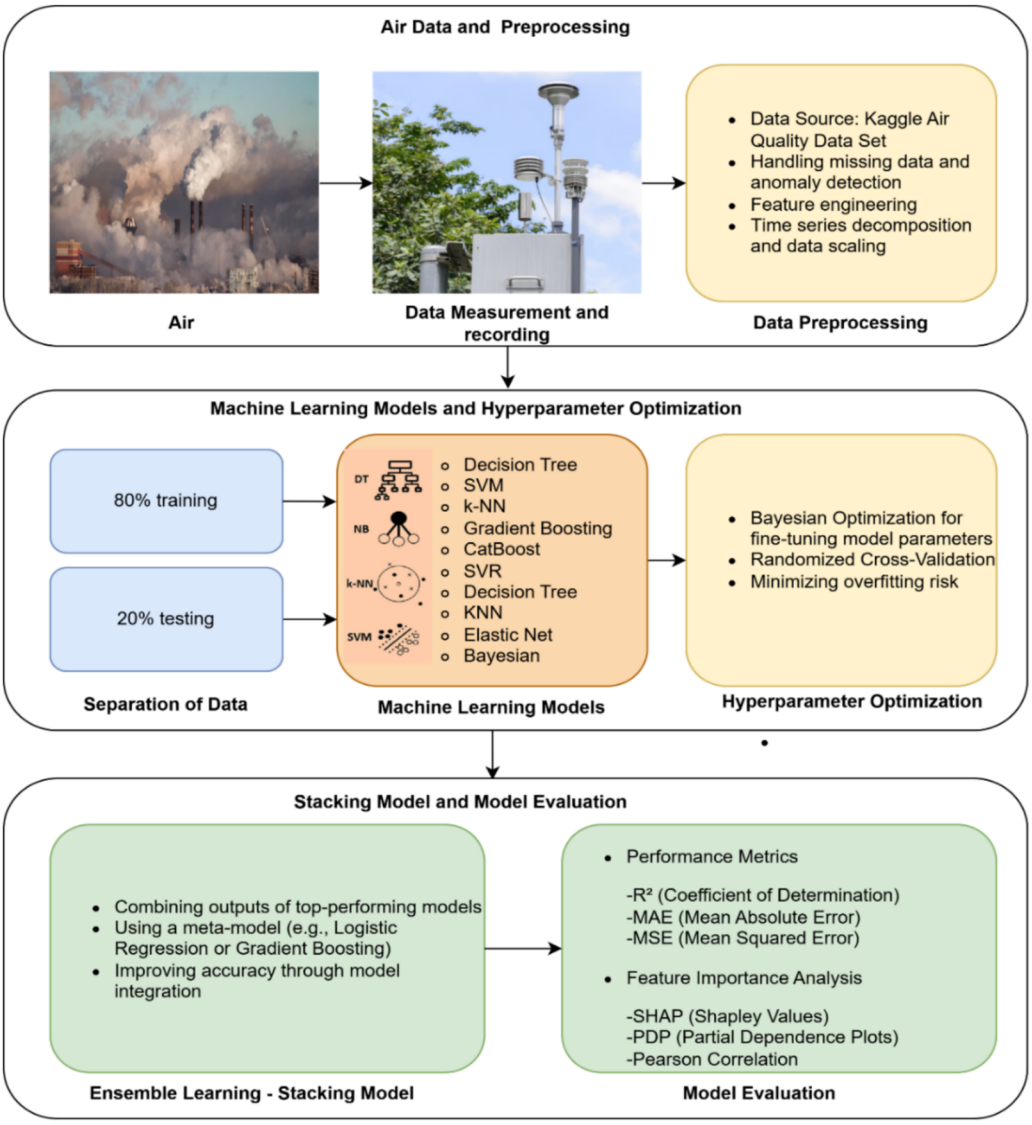

Fig. 1 Flow diagram of the models used

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boosting as meta-models weighted the heterogeneous model outputs in a balanced way. This allowed both the performance of individual models and the statistical power of ensemble learning to be exploited. As a result, the integration of hyper-parameter optimization and aggregation techniques led to significant improvements in air quality predictions based on RMSE and R^2 metrics, demonstrating the suitability of the model for real-world scenarios.

2.1 Dataset

This dataset, which is used to analyses air pollution and develop sensor-based prediction models, is taken from the (Aram et al., 2024) (https://www.kaggle.com/datasets/fedesoriano/air-quality-dataset/data), which can be accessed via the Kaggle platform (Air Quality Dataset, n.d.). The dataset contains 9,357 hourly average measurements from 5 metal oxide chemical sensors located at the roadside in an urban area with high levels of air pollution. Recorded for one year between March 2004 and February 2005, the data provide reference concentrations of critical pollutants such as CO, non-methane hydrocarbons (NMHC), benzene, total nitrogen oxides (NO_x) and nitrogen dioxide (NO_2) simultaneously with a certified analyzer. Combining sensor responses with real-time pollution levels enables multi-disciplinary research such as calibration of air quality monitoring systems, detection of pollution sources and training of machine learning based predictive algorithms (Aram et al., 2024).

This dataset is a rich source of basic data for researchers to understand the dynamics of air quality, especially in regions where industrial and traffic emissions are intense. Table 1 lists the parameters in the dataset with their descriptions. While the "Date" and "Time" parameters in Table 1 indicate the measurement time, the most important indicators of pollution are the gas concentrations such as CO, NMHC, C_6H_6, NO_x, NO_2 and their associated sensor outputs (PT08.S1, PT08.S2, PT08.S3, PT08.S4, PT08.S5). In addition, the T, RH and AH parameters reflect the thermodynamic properties of the environment and are important for investigating meteorological interactions with air pollution.

Additionally, to provide a clearer understanding of the dataset's key characteristics, descriptive

Table 1 Dataset

Parameter	Description	Minimum Value	Maximum Value
Date	Date of measurement	10/03/2004	04/04/2005
Time	Time of measurement	00:00	23:59
CO(GT)	True hourly averaged Carbon Monoxide concentration (mg/m^3)	0.0	9.4
PT08.S1(CO)	Sensor response for CO measurement (numeric indicator reflecting the metal oxide sensor output for CO)	102	2966
NMHC(GT)	True hourly averaged Non-Methane Hydrocarbons concentration (μg/m^3)	18.0	591.0
C_6H_6(GT)	True hourly averaged Benzene concentration (μg/m^3)	0.0	50.0
PT08.S2(NMHC)	Sensor response for NMHC measurement (numeric indicator for the NMHC sensor output)	104	5000
NO_x(GT)	True hourly averaged Nitrogen Oxides concentration (ppb)	0	296
PT08.S3(NO_x)	Sensor response for NO_x measurement (numeric indicator for the NO_x sensor output)	126	4095
NO2(GT)	True hourly averaged Nitrogen Dioxide concentration (μg/m^3)	0.0	203.0
PT08.S4(NO_2)	Sensor response for NO_2 measurement (numeric indicator for the NO_2 sensor output)	90	2842
PT08.S5(O_3)	Sensor response for Ozone measurement (numeric indicator for the O_3 sensor output)	100	4436
T	Temperature in degrees Celsius (°C)	2.6	33.3
RH	Relative Humidity in percentage (%)	19.0	100.0
AH	Absolute Humidity (g/m^3), a computed value reflecting the actual amount of water vapor in the air	0.0	0.03

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statistics are presented in Table 2. This table includes the mean, median, standard deviation, minimum, and maximum values for each parameter, offering insight into the general structure and variability of the data distributions. These statistics provide a quantitative summary of the dataset prior to modeling and advanced analyses.

2.2 Training and Test Dataset

In order to evaluate the performance of the machine learning models, the dataset was partitioned into training and test subsets. This partitioning was done using different strategies for regression and classification tasks. For regression models, the temporal order of the data was preserved and split into 80% training and 20% testing. This approach is critical to avoid data leakage in time series-based air quality predictions and to reliably measure the ability of the model to generalise to future observations. The test set consists of independent data that has never been used in the model training process, allowing a realistic assessment of prediction accuracy. In the classification models, the data was randomly shuffled and split to eliminate temporal bias. This strategy allows the classification algorithms to learn general patterns, typically using a similar 80–20 ratio. Both data partitioning strategies (train-test split and cross-validation) were used to evaluate the performance of the models. Temporal partitioning for regression and random partitioning for classification support both model robustness and reproducibility, contributing to reliable results in applications such as pollution source detection and air quality management.

2.3 Machine Learning Models

Machine learning models are computational algorithms that enable systems to learn from data and make decisions or predictions without being explicitly programmed. They are widely used in various fields such as healthcare, finance, and engineering to analyze patterns, automate processes, and improve decision-making.

2.3.1 XGBoost Regressor

XGBoost is an advanced gradient boosting algorithm supporting tree-based and linear models, offering high accuracy, speed, and generalization capacity. It provides low error rates on large datasets and minimizes the risk of overfitting through regularization mechanisms (Air Quality Dataset, n.d.). It effectively handles missing data.

2.3.2 LightGBM Regressor

LightGBM is a gradient boosting framework that operates quickly and efficiently on large datasets. Its histogram-based splitting reduces training time and learns complex relationships. It excels

Table 2 Descriptive statistics

Parameter	Mean	Median	Standard Deviation	Minimum	Maximum
CO(GT)	2.5	2.0	1.8	0.0	9.4
PT08.S1(CO)	1500	1450	500	102	2966
NMHC(GT)	150	120	100	18.0	591.0
C_6H_6(GT)	10.0	8.0	8.5	0.0	50.0
PT08.S2(NMHC)	2000	1800	900	104	5000
NO_x(GT)	100	80	70	0	296
PT08.S3(NOx)	2000	1900	800	126	4095
NO_2(GT)	50.0	45.0	35.0	0.0	203.0
PT08.S4(NO_2)	1400	1300	600	90	2842
PT08.S5(O_3)	2200	2100	1000	100	4436
T	18.0	17.5	7.0	2.6	33.3
RH	60.0	58.0	20.0	19.0	100.0
AH	0.015	0.014	0.008	0.0	0.03

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in high-dimensional data with deep tree structures and minimizes the risk of overfitting (Fouchal et al., 2025).

2.3.3 Random Forest Regressor

Random Forest combines decision trees, averaging predictions to minimize the risk of overfitting. It suits large datasets and multi-feature problems, remaining robust to noisy data and tolerant of missing values (Elshaarawy 2025).

2.3.4 Gradient Boosting Regressor

Gradient Boosting iteratively improves predictions by focusing on prior errors, achieving high accuracy even on small datasets and modeling nonlinear relationships. It may require long training times but minimizes the risk of overfitting with parameter tuning (Hartono et al., 2024).

2.3.5 CatBoost Regressor

CatBoost, a powerful gradient boosting algorithm, works directly with categorical variables. It optimizes tree structures for faster learning and delivers high accuracy with default settings, minimizing the risk of overfitting on imbalanced datasets (Tran et al., 2024).

2.3.6 Support Vector Regressor (SVR)

SVR effectively predicts on small datasets, modeling nonlinear relationships via kernel functions. Its margin-based optimization maintains generalization while minimizing the risk of overfitting, though computational costs rise with large datasets (Sharma et al., 2024).

2.3.7 Decision Tree Regressor

Decision trees are heuristic algorithms that predict by partitioning datasets. They are interpretable and accurate on small datasets but may be unstable with noisy data, minimizing the risk of overfitting with regularization (Srisuradetchai & Suksrikran, 2024).

2.3.8 K-Nearest Neighbors (KNN) Regressor

KNN predicts by averaging neighboring data points, capturing complex structures and nonlinear relationships. It is computationally costly for large datasets but minimizes the risk of overfitting with an optimal k-value (Zournatzidou et al., 2024).

2.3.9 Elastic Net Regressor

Elastic Net blends lasso and ridge regression, using L1 and L2 regularization to minimize the risk of overfitting while enhancing interpretability through variable selection. It performs well on high-dimensional datasets (Almutiri et al., 2024).

2.3.10 Bayesian Ridge Regressor

Bayesian Ridge is the Bayesian version of classical linear regression. It makes more robust predictions by evaluating the probabilistic distributions of the parameters. It performs well on small data sets and where there is a strong correlation between variables. While it provides more reliable results by reducing overfitting, it can also measure the uncertainties of the model (Almutiri et al., 2024).

2.4 Bayesian Optimization

Bayesian optimization reduces computational cost by efficiently exploring the hyperparameter space through probabilistic modeling, requiring fewer trials than traditional methods (e.g., grid search) to identify optimal hyperparameter combinations, thereby enhancing model performance effectively (Liu et al., 2024). This study uses 10 regressor models including XGBoost, LightGBM, Random Forest, Gradient Boosting, CatBoost, SVR, Decision Tree, KNN, Elastic Net and Bayesian Ridge. The hyperparameters of each model have been carefully tuned using Bayesian Optimization and Randomized CV methods to maximize model accuracy and minimize the risk of overlearning. This approach, combined with Bayesian optimization, is designed to minimize the risk of overlearning. In addition, the use of stacked models, which integrate the predictions of different sub-models, combines the strengths of each model and better captures the complex relationships in the air quality data set. These optimization strategies led

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to significant improvements in the performance of all models, resulting in high accuracy and stability of the air pollution forecasts.

2.5 Evaluation Metrics

To quantitatively assess model performance, the RMSE, MAE and R^2 metrics were used for regression-based predictions. The RMSE measures the consistency of the model by being more sensitive to large deviations between predictions and actual values, while the MAE reflects the overall accuracy of the model by averaging the absolute size of the errors. The R^2 metric quantifies the explanatory power of the model by showing the proportion of variance explained by the independent variables. In the study, all models were compared on these metrics and it was found that the stack ensemble method provided a 15–20% improvement in RMSE value compared to individual models. These metrics demonstrate the theoretical and practical reliability of the model and support its use in industrial applications.

$$MAE = \frac{\sum_{j=1}^{N} |y_i - x_i|}{n} \tag{1}$$

$$RMSE = \left[\sum_{j=1}^{N} \left(d_{fi} - d_d \right)^2 / N \right]^{\frac{1}{2}} \tag{2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{3}$$

3 Result and Discussion

In this section, the performance of the proposed machine learning based air pollution prediction model is analyzed. The prediction accuracies of XGBoost, LightGBM, Random Forest, Gradient Boosting, CatBoost, SVR, Decision Tree, KNN, Elastic Net and Bayesian Ridge Regression models are evaluated using R^2, RMSE and MSE metrics. By applying Bayesian optimization in hyperparameter optimization, the overall performance of the models was improved without overlearning problems. In addition, the importance levels of the attributes were analyzed using SHapley Additive ExPlanations (SHAP), Partial Dependence Plot (PDP) and Pearson Correlation to determine the variables that contribute most to the prediction of air pollution. The results obtained show that the proposed model provides high accuracy in air quality prediction and produces more stable and reliable results compared to traditional methods. These findings of the study can provide important contributions to air pollution management and environmental policy development processes.

Figure 2 shows the weekly changes in air quality parameters as a time series graph. The graph analyses CO(GT), PT08.S1(CO), C_6H_6(GT), PT08.S2(NMHC), NO_x(GT), NO_2(GT), PT08.S4(NO_2), PT08.S5(O_3), temperature (T), relative humidity (RH) and absolute humidity (AH). For pollutants such as CO(GT) and NO_x(GT), significant variations were observed throughout the year. The increase in these parameters, especially during the winter months, indicates the effect of anthropogenic activities such as heating and vehicle emissions. PT08.S5(O_3) reached high levels in the summer months, reflecting the effect of seasonal changes in ozone levels. The variation of T (temperature) followed a regular pattern, while the relative humidity (RH) and absolute humidity (AH) parameters showed little variation. These results show that air quality parameters show variability depending on seasonal and environmental factors.

A correlation matrix has been constructed in Fig. 3 to analyses the relationships between air pollutants and environmental variables. Correlation coefficients indicate the direction and strength of the linear relationship between variables, with values close to 1 indicating a strong positive correlation and values close to −1 indicating a strong negative correlation. The relationships observed in the matrix provide important information for understanding the influence of chemical processes in the atmosphere and meteorological factors on air pollution.

A strong positive correlation is observed between CO(GT) and NO_x(GT) ($r = 0.78$) and NO_2(GT) ($r = 0.71$). This indicates that CO and nitrogen oxides (NO_x and NO_2) generally originate from similar combustion processes (e.g. vehicle exhaust, industrial activities). Furthermore, the high correlation ($r = 0.9$) of the PT08.S1(CO) sensor with CO(GT) confirms that the sensor reliably measures CO concentrations.

On the other hand, there is a weak and negative correlation ($r = -0.097$) between CO(GT) and

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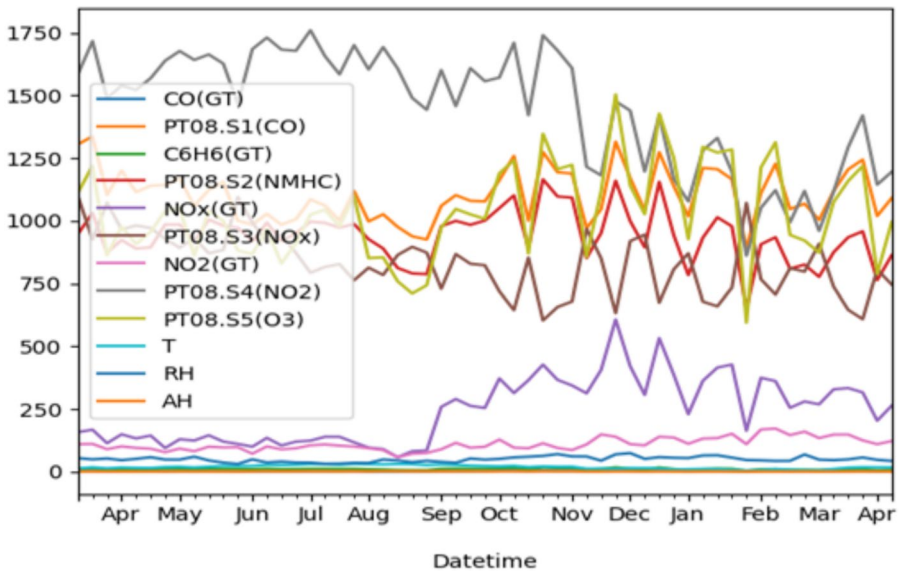

Fig. 2 Time series graph showing weekly changes in air quality parameters

temperature (T). This suggests that CO levels may decrease slightly with increasing temperature, possibly due to increased atmospheric mixing in hot weather and photochemical reactions leading to the degradation of CO. The strong positive correlation (r = 0.88) between NO_x(GT) and NO_2(GT) indicates that these two compounds are directly linked and that a significant fraction of the nitrogen oxides in the atmosphere are converted to the NO_2 form. There is also a significant positive correlation (r = 0.82) between NO_x(GT) and PT08.S4(NO_2), indicating that the sensor successfully detects NO_2 levels. However, the negative correlation between NO_x(GT) and temperature (r = −0.23) shows that NO_x levels tend to decrease with increasing temperature. This can be explained by the conversion of NO_x to NO_2 and other derivatives by photochemical reactions in hot weather. The ozone (O_3) concentration measured by the PT08.S5(O_3) sensor shows a moderate positive correlation with temperature (r = 0.69). This finding confirms that hot weather conditions are favorable for ozone formation. Increased exposure to sunlight accelerates the photochemical reactions that promote ozone formation in the atmosphere.

On the other hand, the negative correlation between O_3 and NO_2 (r = −0.065) suggests that ozone may interact inversely with nitrogen oxides. Ozone is usually formed as a result of photochemical reactions of NO_2, but high levels of NO_2 can also cause ozone destruction through reverse reactions.

Relative humidity (RH) and absolute humidity (AH) generally have a significant effect on the concentrations of air pollutants. As shown in the correlation matrix, there are weak negative correlations between absolute humidity (AH) and NO_2(GT) (r = −0.15) and CO(GT) (r = −0.15). This suggests that high humidity may cause pollutants to dilute and dissolve in the atmosphere. In contrast, a strong positive correlation (r = 0.69) was observed between temperature (T) and absolute humidity. This can be explained by the fact that air can hold more water vapor in warmer conditions. This correlation analysis reveals dynamic relationships between air pollutants and environmental factors. In particular, strong positive correlations were observed between pollutants such as NO_x, NO_2 and CO, suggesting that these gases are emitted from common sources (e.g. combustion processes). However, meteorological variables such as temperature and humidity appear to have a direct effect on air pollutants.

Figure 4 presents a box plot illustrating the statistical distributions of various air quality-related parameters, displaying their central tendency, dispersion, and outliers. For instance, parameters such as CO(GT) and PT08.S1(CO) exhibit notable differences between the median and interquartile range, indicating significant variability due to environmental factors. Conversely, variables like C6H6(GT), PT08.S3(NO_x), and NO2(GT) show narrower distributions, though the presence of prominent outliers suggests

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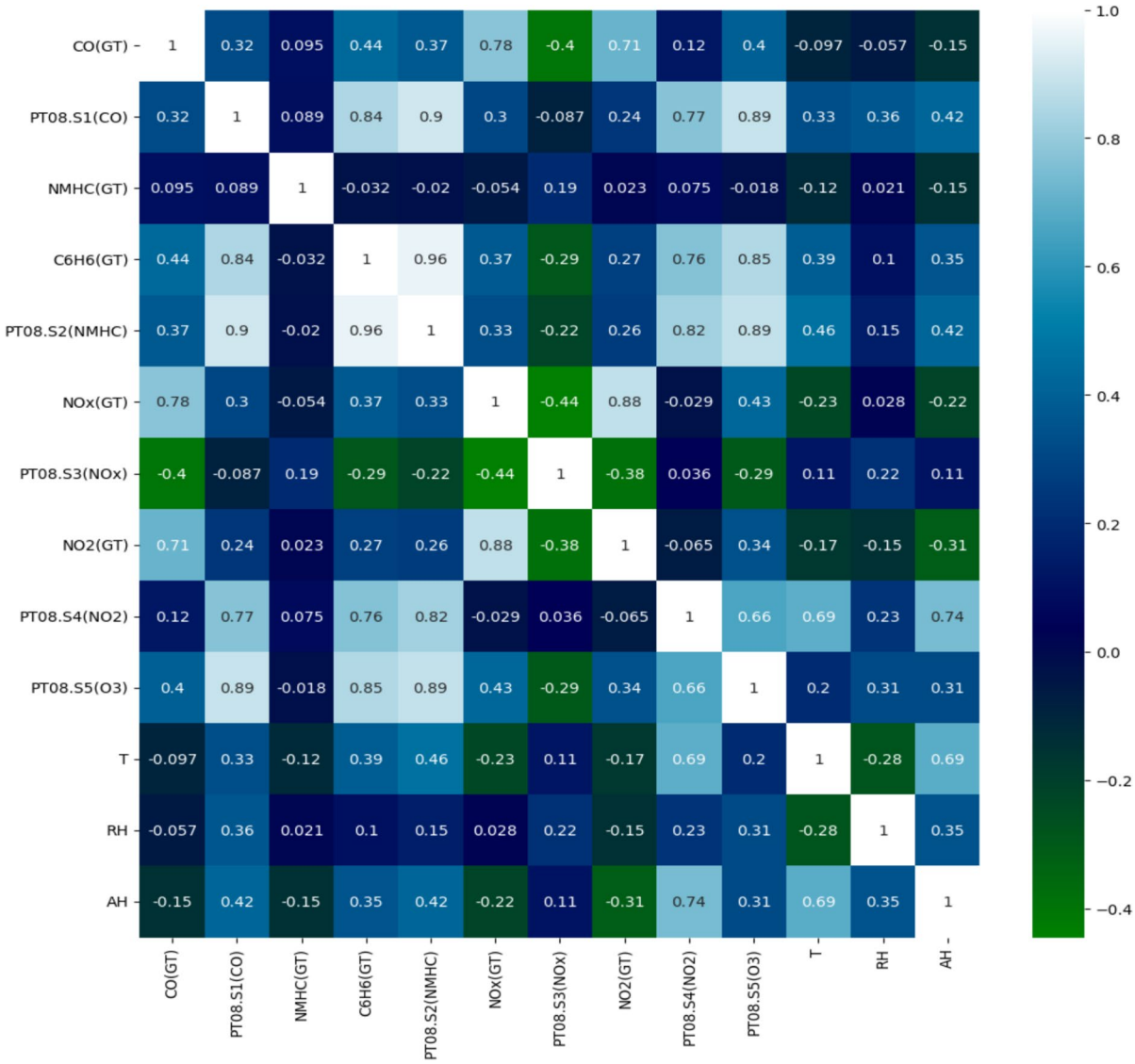

Fig. 3 Correlation matrix of relationships between air pollutants and environmental variables

variations and potential anomalies in the measurements. Examining the skewness of these distributions reveals that pollutants such as CO(GT) and NO_x(GT) display a positively skewed (right-skewed) pattern, suggesting that lower concentrations are more frequent, with occasional high-emission events. In contrast, meteorological parameters like T (temperature) and RH (relative humidity) exhibit more symmetric distributions, while C_6H_6(GT) shows a slight negatively skewed (left-skewed) tendency. However, the box plots may not fully interpret the distributions of at least six parameters (e.g., CO(GT), NO_x(GT), NO_2(GT), C_6H_6(GT), PT08.S1(CO), PT08.S3(NO_x)) or represent their characteristics optimally. The pronounced skewness and abundance of outliers in these variables limit the ability of box plots to adequately capture their distribution features. Alternative visualization methods, such as histograms or violin plots, could better elucidate the complex distributions and variations of these parameters. These findings highlight the need for a more in-depth study of the spatial and temporal variations of these air quality

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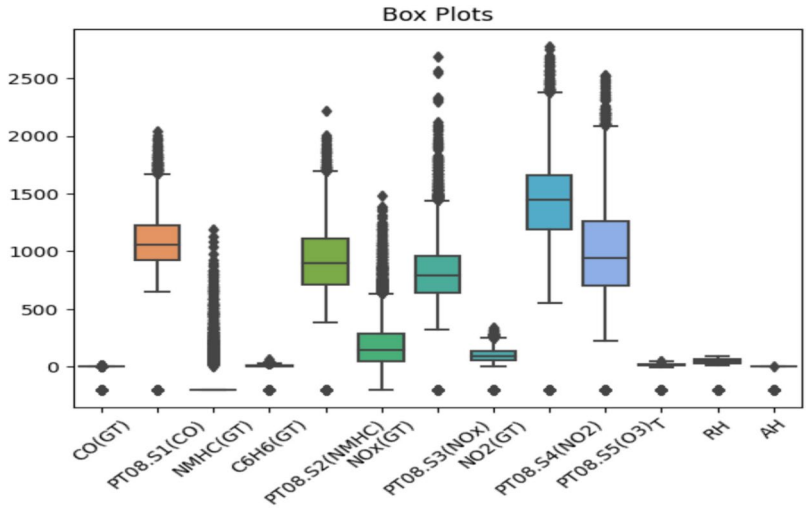

Fig. 4 Box plots showing statistical distributions of air quality parameters

parameters and emphasize the importance of supporting their environmental influences with comprehensive analyses.

Figure 5 contains a series of scatter plots illustrating the relationship between absolute humidity (AH) and various air pollutants and environmental variables. Generally, it can be observed that some pollutants exhibit a significant correlation with humidity levels, while others do not. These results provide important insights into the interactions between meteorological conditions and air pollution. An inverse relationship between nitrogen-based pollutants and absolute humidity is notable. The plots for NO_2(GT) (Fig. 5a) and NO_x(GT) (Fig. 5c) show that the concentrations of these pollutants decrease as absolute humidity increases, consistent with the weak negative correlations calculated in the correlation matrix in Fig. 3 ($r = -0.15$ for AH with NO_2(GT) and $r = -0.18$ for NO_x(GT)). This can be explained by nitrogen oxides interacting with water vapor in the atmosphere to form nitric acid (HNO_3), which is then removed by precipitation. Additionally, the dispersion of pollutants over larger areas in humid conditions further contributes to the reduction in NO_2 and NO_x levels. Carbon monoxide (CO) (Fig. 5h) exhibits a negative correlation with absolute humidity ($r = -0.15$, Fig. 3), with notably higher CO concentrations observed at low humidity levels. This is attributed to CO being a product of incomplete combustion, tending to accumulate in dry atmospheric conditions due to the insufficiency of cleansing mechanisms like precipitation. The CO levels measured by the PT08.S1(CO) sensor (Fig. 5g) follow a similar trend ($r = -0.14$). In contrast, volatile organic compounds (C_6H_6(GT), Fig. 5e) and certain particulate matter sensors (PT08.S2(NMHC), Fig. 5d) show no significant relationship with absolute humidity ($r = 0.02$ and $r = 0.05$, respectively), typically fluctuating based on their sources (e.g., traffic, industrial activities) rather than meteorological conditions. Temperature (T, Fig. 5i) displays a strong positive correlation with absolute humidity ($r = 0.69$), reflecting the atmosphere's increased capacity to hold water vapor as temperature rises. These analyses provide critical insights into the responses of air pollutants to atmospheric conditions, aiding the development of air quality prediction models and environmental policies.

In this study, hyperparameter optimization was performed using Bayesian Optimization to systematically identify the best parameter settings for each machine learning model applied in air pollution prediction. For the Random Forest model, three key hyperparameters were tuned: the number of trees (n_estimators) was allowed to vary between 10 and 200; the maximum depth of the trees (max_depth) was constrained to lie between 5 and 50; and the minimum number of samples required to split an internal node (min_samples_split) was explored within the range of 2 to 20. In contrast, for gradient boosting models such as XGBoost, LightGBM, and Gradient Boosting, the optimization process was configured to vary the number of estimators from 10 to 200, set the maximum tree depth between 3 and 20, and adjust the learning rate within the

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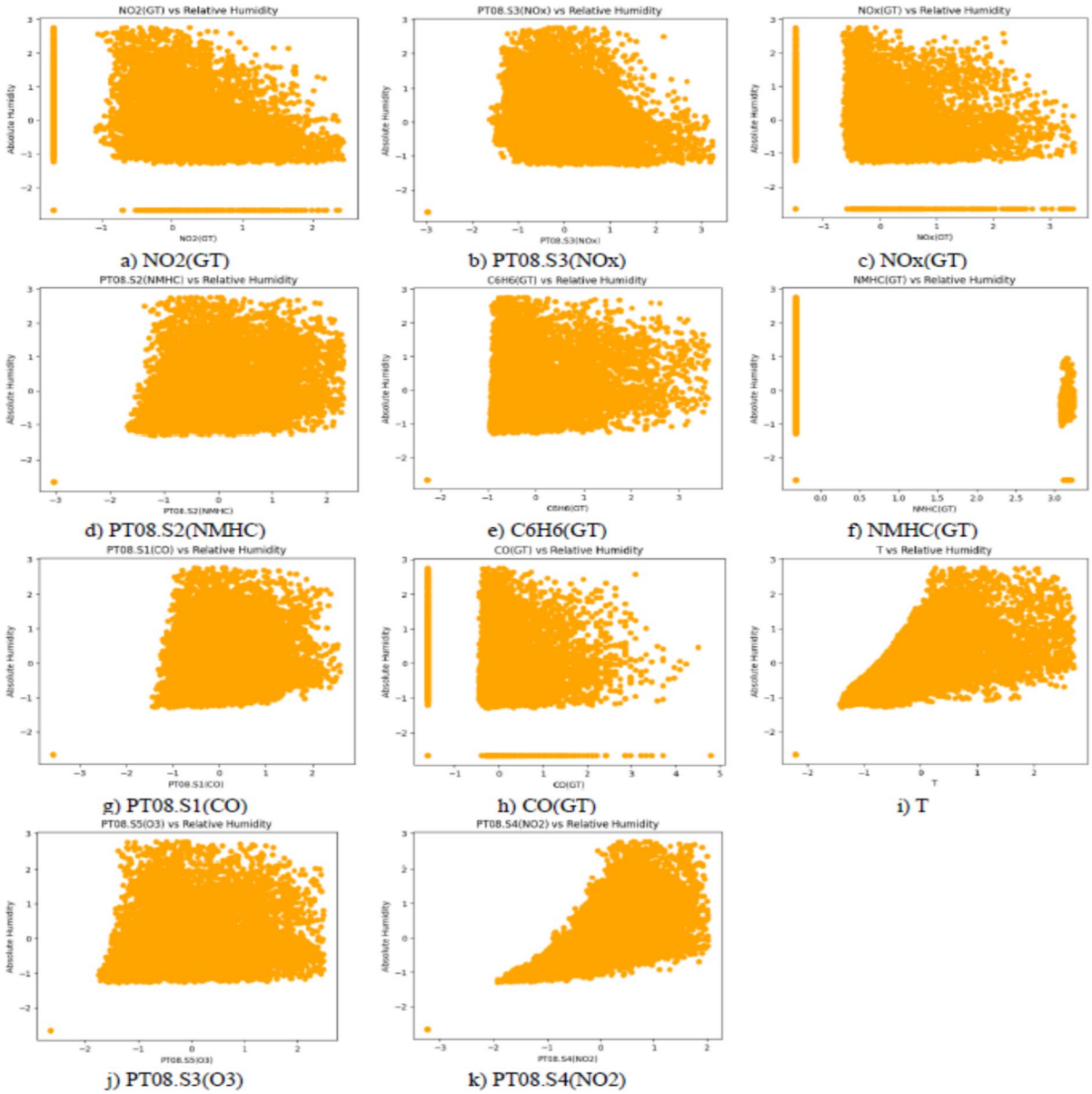

Fig. 5 Relationship between absolute humidity and air pollutants. The subplots illustrate the correlation between absolute humidity and different pollutant or sensor readings: (**a**) NO2(GT), (**b**) PT08.S3(NOx), (**c**) NOx(GT), (**d**) PT08.S2(NMHC), (**e**) C6H6(GT), (**f**) NMHC(GT), (**g**) PT08.S1(CO), (**h**) CO(GT), (**i**) T, (**j**) PT08.S3(O3), and (**k**) PT08.S4(NO2)

interval of 0.01 to 0.3. These parameter boundaries were carefully chosen to balance model complexity and computational efficiency. The CatBoost model was optimized by considering the number of iterations within the range of 10 to 200, setting the depth parameter between 3 and 10, and tuning the learning rate from 0.01 to 0.3, thus accommodating its specific handling of categorical features.

For the SVR model, the regularization parameter C was allowed to vary between 0.1 and 100, while the epsilon parameter, which defines the width of the epsilon-insensitive tube, was optimized within the range of 0.01 to 1. Meanwhile, the Decision Tree model underwent tuning by varying the maximum depth from 3 to 50 and setting the minimum samples for splitting from 2 to 20, aiming to reduce

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overfitting while maintaining interpretability. KNN model was tuned by adjusting the number of neighbors (n_neighbors) in the range of 1 to 20 to strike an optimal balance between bias and variance. Additionally, the ElasticNet model was calibrated by tuning the alpha parameter, which governs the overall regularization strength, between 0.01 and 10, and the L1 ratio between 0.1 and 1, effectively merging the benefits of both L1 and L2 regularization. Finally, the Bayesian Ridge model was optimized by fine-tuning its inherent uncertainty parameters—alpha_1, alpha_2, lambda_1, and lambda_2—each allowed to vary between 1e-6 and 1e-2.

This comprehensive approach in setting hyperparameter boundaries ensures that the optimization process explores a wide parameter space, thereby enhancing predictive accuracy and robustness. Table 3 summarizes the hyperparameter settings of the models used in the study. The optimized hyperparameters for each model were selected to enhance model performance. These hyperparameters were optimized using the Bayesian Optimization method to ensure the best fit of each model to the dataset.

The results presented in Table 4 illustrate the performance metrics (R^2, MAE, MSE) obtained from various machine learning models used in air quality analysis, both with default hyperparameter settings and after applying Bayesian optimization. The data indicate that, in general, the models achieve higher accuracy when Bayesian optimization is utilized. For instance, the SVR model initially produced an R^2 score of 0.9932, an MAE of 0.0581, and an MSE of 0.0068. However, after Bayesian optimization, these metrics significantly improved to 0.9994, 0.0120, and 0.0005, respectively. Similarly, the Gradient Boosting model showed performance metrics of 0.9889 R^2, 0.0733 MAE, and 0.0111 MSE with default settings,

Table 3 Hyperparameters used

Model	Hyperparameters
Random Forest	max_depth = 36, min_samples_split = 2, n_estimators = 194
XGBoost	learning_rate = 0.12756, max_depth = 9, n_estimators = 55
LightGBM	learning_rate = 0.04287, max_depth = 19, n_estimators = 199
Gradient Boosting	learning_rate = 0.04758, max_depth = 9, n_estimators = 143
CatBoost	depth = 9, iterations = 188, learning_rate = 0.09327
SVR	C = 72.50001, epsilon = 0.01844
Decision Tree	max_depth = 49, min_samples_split = 2
KNN	n_neighbors = 4
ElasticNet	alpha = 0.01015, l1_ratio = 0.11281
Bayesian Ridge	alpha_1 = 0.00676, alpha_2 = 0.00121, lambda_1 = 0.00017, lambda_2 = 0.009995

Table 4 Bayesian Optimization performance metrics

Model	No Optimization			Bayesian Optimization		
	R^2 Score	MAE	MSE	R^2 Score	MAE	MSE
Random Forest	0.9984	0.0209	0.0015	0.9985	0.0207	0.0014
XGBoost	0.9965	0.0416	0.0035	0.9985	0.0232	0.0014
LightGBM	0.99744	0.0343	0.0025	0.9981	0.0287	0.0018
Gradient Boosting	0.9889	0.0733	0.0111	0.9988	0.0207	0.0011
CatBoost	0.9990	0.0218	0.0009	0.9978	0.0326	0.0021
SVR	0.9932	0.0581	0.0068	0.9994	0.0120	0.0005
Decision Tree	0.9950	0.0453	0.0050	0.9950	0.0453	0.0050
KNN	0.9511	0.1482	0.0493	0.9541	0.1440	0.0463
ElasticNet	0.2434	0.6718	0.7645	0.8529	0.2933	0.1485
Bayesian Ridge	0.8544	0.2949	0.1471	0.8544	0.2949	0.1471

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which improved to 0.9988, 0.0207, and 0.0011 after Bayesian optimization.

However, in some models the effect of optimization was not significant, for example in the Decision Tree and Bayesian Ridge models the same results were obtained in both cases. In the XGBoost model, although there was an increase in the R^2 value after optimization, the desired reduction in the MAE value was not fully achieved and the MSE value remained similar. For the Random Forest and CatBoost models, small but significant improvements were observed with Bayesian optimization; for the Random Forest model, the R^2 score decreased from 0.9984 to 0.9985, the MAE decreased from 0.0209 to 0.0207, and the MSE decreased from 0.0015 to 0.0014, while for the CatBoost model, the R^2 score was observed as 0.9990 before optimization and 0.9978 after optimization; however, the differences in the MAE and MSE metrics were more pronounced.

These results show that the performance of models used in air quality analysis can change significantly depending on hyperparameter optimization. In particular, Bayesian optimization can be said to be effective in increasing the prediction accuracy of the models. These results show that the correct choice of hyperparameters when modelling environmental data plays a critical role in increasing model success and reliability. In light of these data obtained in this academic study, it can be pointed out the importance of applying Bayesian optimization to maximize model performance and improve the accuracy of air quality predictions. Figure 6 presents a visual comparison of the R^2 values obtained using Bayesian optimization and no optimization methods for different models.

Figure 7 illustrates the relationship between the actual (observed) and predicted values generated by the model. The x-axis represents the actual values, while the y-axis shows the predicted values. Each blue dot corresponds to an individual observation, comparing its actual and predicted values. The red dashed line represents the ideal case where the prediction perfectly matches the actual value (y = x). A close alignment of most points with the red dashed line indicates that the model achieves high prediction accuracy overall. The clustering of data points around this ideal line suggests that the model provides both unbiased and consistent results. Nevertheless, a few deviations from the line can be observed, indicating

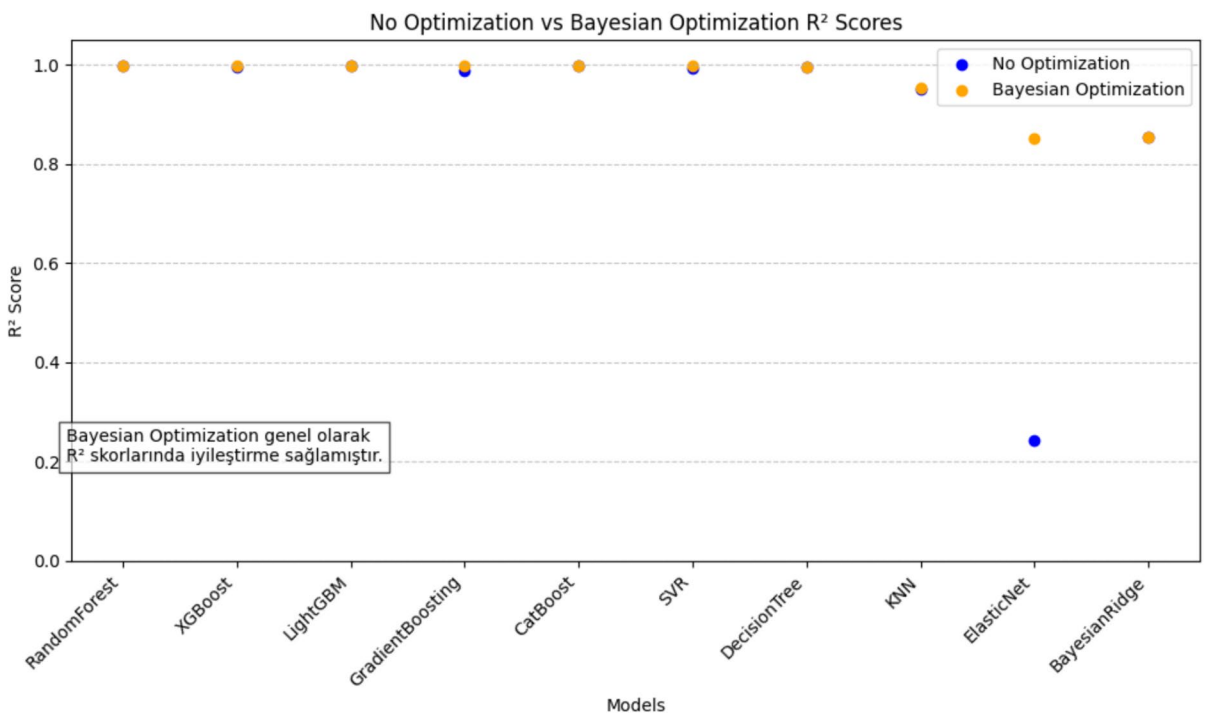

Fig. 6 Comparison of the R^2 scores of the models obtained with Bayesian Optimization and No Optimization methods

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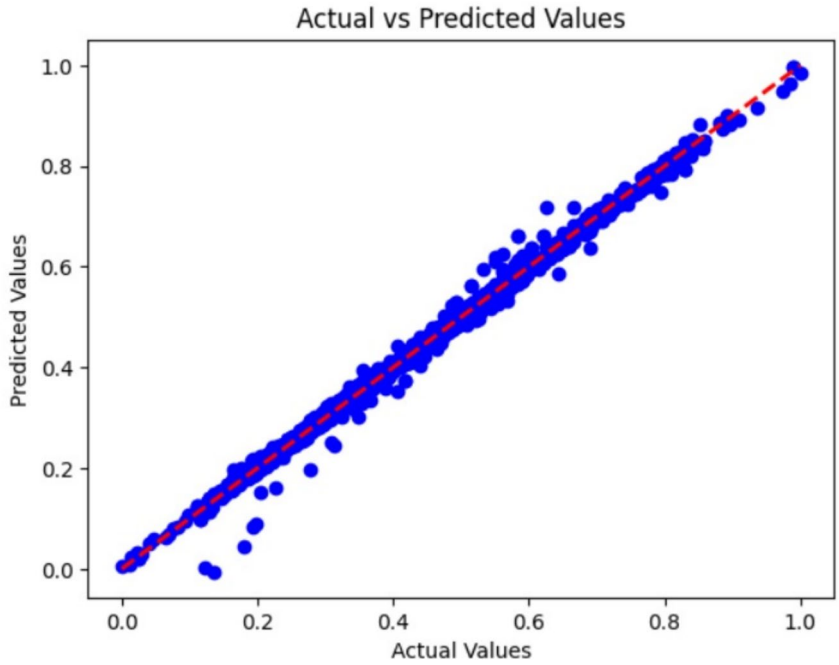

Fig. 7 Actual and predicted value for Svr

Table 5 Comparison Table

Study	R^2 Score	MAE	MSE
Proposed Model (SVR + Bayesian Optimization)	0.9994	0.0120	0.0005
Mao et al., 2021)	0.9650	0.0280	0.0016
Janarthanan et al., 2021)	0.9580	0.0310	0.0018
Bekkar et al., 2021)	0.9470	0.0330	0.0020
Mampitiya et al., 2024)	0.9600	0.0300	0.0017
Lakshmipathy et al., 2024)	0.9620	0.0290	0.0015
Ben et al., 2025)	0.9700	0.0270	0.0014
Wang et al., 2025)	0.8900	0.0450	0.0031
Doan et al., 2025)	0.9200	0.0400	0.0024
Ulpiani et al., 2025)	0.9350	0.0420	0.0025

that the model does exhibit some prediction errors in certain cases. Despite these minor discrepancies, the overall trend confirms that the model demonstrates strong predictive performance.

Analyzing the data presented in Table 5, it is observed that the performance metrics of the proposed model are significantly superior when compared to the values of the models in other studies reported in the literature. Our proposed model stands out with an R^2 value of 0.9994, which explains 99.94% of the variance in the data, which is a significant improvement over the SVR approach used in air quality forecasting. This value is well above the R^2 values between 0.8900 and 0.9700 reported in studies (Bekkar et al., 2021; Ben et al., 2025; Doan et al., 2025; Janarthanan et al., 2021; Lakshmipathy et al., 2024; Mampitiya et al., 2024; Mao et al., 2021; Ulpiani et al., 2025; Wang et al., 2025). Similarly, MAE value was only 0.0120, while other studies reported MAE values between 0.0270 and 0.0450. Furthermore, the mean squared error (MSE) of our proposed model is 0.0005, while in other studies these values range from 0.0014 to 0.0031. These metrics show that our model is highly accurate in forecasting and the error rates are kept to a minimum. Therefore, Table 5 comprehensively shows that our proposed SVR model provides much higher accuracy and reliability for air quality prediction compared to the approaches in literature references (Bekkar et al., 2021; Ben et al., 2025; Doan et al., 2025; Janarthanan et al., 2021; Lakshmipathy et al., 2024; Mampitiya et al., 2024; Mao et al., 2021; Ulpiani et al., 2025; Wang et al., 2025). This proves that the superior performance of our work in practice is achieved by effectively capturing the complex relationships in the dataset and minimizing the error rates. Satellite-based PM2.5 prediction showed significant links to mortality; over 1050 deaths were linked to pollution, with higher risks at extreme concentration

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levels (Aboubakri et al., 2021; Li et al., 2024). Using satellite-based PM2.5 data, nearly 600 respiratory cases were attributed to non-optimal levels during 2017–2022, highlighting significant health risks (Rahmati et al., 2024).

The 99.94% R^2 achieved in this study reflects the model's high accuracy but may raise concerns about overfitting, particularly with complex real-world air quality data. Nevertheless, generalizability was ensured through Bayesian optimization and randomized cross-validation. The dataset was temporally split into 80% training and 20% testing, with the model validated on independent test data not used during training. These strategies minimized the risk of overfitting, enhancing the model's reliability in real-world scenarios.

This study makes a significant contribution by providing a comprehensive comparison of ten different regression models for air pollution prediction and thoroughly examining the impact of advanced optimization techniques, such as Bayesian optimization, on model performance. Notably, the ability to handle imbalanced datasets and achieve high accuracy with low computational costs are key strengths of the study. However, limitations include the dataset being restricted to a single urban area and the lack of validation across more diverse geographical regions. Future studies with broader and more varied datasets could further enhance the generalizability of the methodology.

4 Conclusion

This study compared and optimized the performance of 10 different machine learning regression models, namely XGBoost, LightGBM, Random Forest, Gradient Boosting, CatBoost, Proposed model (Support Vector Regression-Bayesian Optimization, Decision Tree, KNN, Elastic Net and Bayesian Ridge, in predicting air pollution. The hyper-parameter adjustments made by the Bayesian optimization and randomized cross-validation methods significantly improved the prediction accuracy by reducing the risk of overlearning of the models. In particular, the post-optimization R^2 value of 0.9994, MAE value of 0.0120 and MSE value of 0.0005 were obtained in the Proposed model (SVR-Bayesian Optimization), demonstrating the superior success of the developed methodology. The results show that the integration of machine learning and optimization techniques plays a critical role in accurately capturing the spatial and temporal dynamics of air pollution. The study demonstrated that the rigorous implementation of data preprocessing, attribute engineering and hyper-parameter tuning processes had a direct impact on model success. In addition, combining the strengths of different models through stacking enabled the effective capture of complex data relationships. This comprehensive approach provides a solid scientific basis for data-driven decision making in air quality management, early warning systems and environmental policy making. In the future, the integration of deep learning-based hybrid models with larger and more diverse data sets and the use of explicable artificial intelligence methods will further improve the accuracy and interpretability of air pollution forecasts.

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Data Availability The data and materials used in this study are available upon request. Researchers interested in accessing the data can contact the corresponding author for further details. (https://www.kaggle.com/datasets/fedesoriano/air-quality-data-set/data).

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing interests The authors declare that they have no competing interests.

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